

Cognitive Modeling Using Artificial Intelligence

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Keywords

cognitive modeling, artificial intelligence, psychological theory

Abstract

Recent progress in artificial intelligence (AI) is exciting, but can AI models tell us about the human mind? AI models have a long history of being used as theoretical artifacts in cognitive science, but one key difference in the current generation of models is that they are stimulus computable, meaning that they can operate over stimuli that are similar to those experienced by people. This advance creates important opportunities for deepening our understanding of the human mind. We argue here that the most exciting of these is the use of AI models as cognitive models, wherein they are trained using human-scale input data and evaluated using careful experimental probes. Such cognitive models constitute a substantial advance that can inform theories of human intelligence by helping to explain and predict behavior.

Contents

INTRODUCTION	544
MODELS OF MIND	546
Frameworks, Theories, and Models	546
Model–Mechanism Mappings	547
Stimulus Computable Models	549
Modeling Naïve Psychology Using Large Language Models	549
MAPPING BETWEEN LEARNING MODELS AND HUMAN SYSTEMS	550
Model Training	550
Model Evaluation	552
OTHER THEORETICAL CONTRIBUTIONS OF AI MODELS	554
Machine Research Assistants	554
Human Simulators	555
Other Minds	556
CHALLENGES AND OPPORTUNITIES	557
Transparency and Reproducibility	557
Population Generalizability	558
Interpretability	558
CONCLUSION	559

INTRODUCTION

The promise of cognitive science is to provide an explanatory and predictive understanding of the behavior of minds. Since its inception, cognitive science also has been concerned with the intersection between natural and artificial intelligence (AI). For example, an editorial in the first issue of the journal *Cognitive Science* ends noting the “excitement surrounding the paradigm shift in the study of cognition made possible by the synthesis of artificial intelligence, psychology, and linguistics” (Collins 1977, p. 2). This synthesis has only accelerated in the 48 years since. In this review, we explore the theoretical opportunities afforded by new AI models. We argue that these can be used as models of mind—scientific artifacts to predict and explain human cognition—especially when they are trained using human-like input data and evaluated using experimental probes parallel to those used in human experiments.

Recent years have brought immense progress in AI as an independent field of study (Goodfellow et al. 2016). One important milestone was the revival of deep neural networks and the striking improvements in image recognition that they yielded when combined with a large data set of labeled images (Krizhevsky et al. 2012). This milestone was soon followed by remarkable successes in reinforcement learning and unsupervised language modeling (Brown et al. 2020, Silver et al. 2016). Public attention was captured especially by the launch of ChatGPT, an interactive dialogue interface to a predictive language model, in late 2022. AI is becoming integrated into the broader software ecosystem in which we increasingly live, mediated by smartphones, computers, tablets, watches, glasses, and other devices.

It is not immediately obvious that advances in AI as a product—or even as a technical artifact—will yield scientific advances in the understanding of the mind. The same editorial in the first issue of *Cognitive Science* warns against submitting “descriptions of intelligent systems that do not provide any insight into human processing” (Collins 1977, p. 2). And yet modern AI models exhibit

MODERN AI MODELS

Modern AI models are typically deep neural networks: sets of simulated neurons connected to each other through learnable weights (Goodfellow et al. 2016, LeCun et al. 2015). The units of these networks diverge substantially from biological neurons, which have more complex temporal dynamics and different learning rules (Marblestone et al. 2016). Instead, neural networks consist of millions or billions of units whose connection weights can be adjusted to optimize a given learning objective.

Large language models (LLMs) have at their core a simple objective: next word prediction. That is, given a string of words $w_1 \dots w_{n-1}$, their learning objective is to make the most precise prediction about w_n . The astonishing observation about LLMs is that complex behaviors emerge when this objective is applied to train very large models on huge amounts of data (Brown et al. 2020).

The vast majority of the most performant AI models are based on the transformer architecture (Vaswani et al. 2017), which uses a mechanism known as attention to learn what aspects of a sequence to upweight to be most successful in optimizing its objective. During pretraining (initial training on data), the neural network's weights are adjusted (typically via backpropagation) to minimize the learning objective. In many systems, subsequent fine tuning can be performed to maximize the network's performance on a particular task (e.g., producing helpful answers to questions).

so many human-like behaviors that their potential implications for scientific understanding loom large over cognitive science.

With the remarkable growth in large language models (LLMs) in particular (see the sidebar titled Modern AI Models), there has been an explosion both of cognitive science investigations of these models and of debate about whether and how these models can be informative about the human mind (Blank 2023, McGrath et al. 2024, Ong 2024, Shah & Varma 2025, van Rooij et al. 2024). This debate has been marked by extreme positions, from the suggestion that AI models solve long-standing puzzles about human cognition (Contreras Kallens et al. 2023, Piantadosi 2024) to the counterclaim that these models are wholly uninformative about cognition (Cuskey et al. 2024, Dupre 2021, Katzir 2023). Many discussions of the utility of AI echo themes of previous debates about the role of computational models in cognitive science as well as discussions of the role of scientific models more generally.

We argue here that modern AI models provide a remarkable theoretical opportunity for psychology and cognitive science (as well as neuroscience, though discussion of these applications is outside our scope). The theoretical use of these models is not fundamentally different from the use of previous computational formalisms such as the first generation of neural network models (e.g., Elman et al. 1996) or, more recently, Bayesian models (Griffiths et al. 2024). Yet the increased capabilities of new models open up exciting possibilities.

Previous eras of modeling have been forced to schematize their inputs or restrict their scope to the simplest perceptual inputs. In contrast, modern AI models are stimulus computable, meaning that they can operate over text, audio, images, and sometimes even videos—that is, rich representations of the perceptual inputs to humans (Carvalho & Lampinen 2025). Also, these models can be evaluated both by their behaviors—for example, their generation of text or other modalities—and through readouts of their internal states via mechanistic interpretability tools. These developments create new opportunities for the alignment of models and human systems of interest such as language processing, visual perception, and reasoning.

In this review, we describe the potential for using stimulus computable models as models of mind—scientific artifacts to help us understand cognition. Our primary focus is on LLMs, but

the points we make generally apply to vision-language models and vision models as well. Cognitive models are only as good as their mapping to the human system of interest, and so we discuss assumptions underlying model training and evaluation. We contrast the cognitive modeling approach with alternative uses of AI models for advancing our theoretical understanding, including their use as machine research assistants, human behavior simulators, and general reference points for the nature of intelligence. We end by considering some limitations of the current models that point the way toward future theoretical opportunities.

MODELS OF MIND

Throughout the sciences, mathematical tools are used to create idealized representations of physical systems [Frigg & Hartmann 2024, Wigner 1990 (1960)]. Often the goal is a statement of how different entities interact quantitatively. For example, in classical mechanics, a set of equations are written that allow their users to make predictions about the behaviors of physical systems, subject to some constraints (for example, on the scale of the objects being predicted).

Theories are then instantiated in models, artifacts that can be manipulated.¹ Scientific models can be computational artifacts in which the manipulations are via simulation, but they can also be physical artifacts (think of a computer simulation of an object falling or a toy car in a small wind tunnel). They create a causally accessible version of a system that can be perturbed—for example, by modifying the weight of the object in the simulation or the design of the toy car. They also allow explanations of the behavior of systems. For example, classical mechanics as a theory allows us to create a model of a system, say, with a point mass falling on a rope. This type of model allows for explanations, letting us say, for instance, that the rope broke because of the amount of force the falling climber exerted on it; had the fall been shorter or the climber lighter, it would have held (Cartwright 1983). How does this process play out in psychology and cognitive science?

Frameworks, Theories, and Models

In psychology and cognitive science, the situation is more complicated than in the physical sciences: While the ultimate goal is still prediction and explanation, the set of candidate entities have been more variable. Many different things get called theories in psychology, and many different kinds of things get called models. These practices can lead to considerable confusion (Frank et al. 2025). In some cases, the focus of theory development is simply defining the particular entities (known as constructs) whose relations are to be studied (Cronbach & Meehl 1955). For example, one of the most central debates in psychology is whether intelligence is a single construct or not (Van der Maas et al. 2014).

We use the term “frameworks” to describe general ways of looking at a domain or problem that do not themselves lead to testable predictions (Kuhn 1970). An example of a framework might be Bronfenbrenner’s ecological systems theory, which describes the interconnectedness of many aspects of children’s development, from their family context to their school, neighborhood, and culture (Bronfenbrenner 1992). Although it is called a theory, it is really a framework, since it does not make specific empirical predictions. It certainly cannot be falsified in the Popperian sense: There are no observations that would convince a developmental scientist that a child’s development is not embedded in context.

In contrast, a theory describes a set of causal relationships between (possibly unobserved) psychological constructs. Verbal theories may simply define directional relationships between

¹We’ve been talking about AI models, which just means something like AI artifacts: systems that can be trained and evaluated on data. However, now we need to overload the term by talking about scientific models.

variables—for example, that more responsive caregiving should lead to more secure attachment between the child and the caregiver. Cashing this theory out into predictions about observable data requires a substantial number of further (theory-laden) definitions: We need to know what counts as responsive caregiving and how we measure secure attachment. In practice, in psychology, this often means creating an operationalization—a means to measure each of these variables within a particular study. For at least 50 years there have been calls for these verbal theories to be translated into formal or computational models that provide the kind of prediction and explanation afforded by physical theories (Frankenhuis et al. 2023, Guest & Martin 2021, Harris 1976, Smaldino 2020).

A model in psychology and cognitive science is an artifact that represents a target system, for example, visual working memory, word learning, or intuitive psychology.² These models are almost always either simple formal descriptions (i.e., equations) or computational artifacts. These artifacts take inputs and perform some computation to produce an output. As with models throughout the sciences, they are models by virtue of their correspondence with the system of interest: the modeler selects aspects of a target system to represent as well as, often implicitly, what aspects of the target system not to represent. (Figure 1).

Model–Mechanism Mappings

Psychological systems can rarely be described via one or two simple equations; typically psychological processes operate over complex stimuli that the mind perceives or processes. Thus,

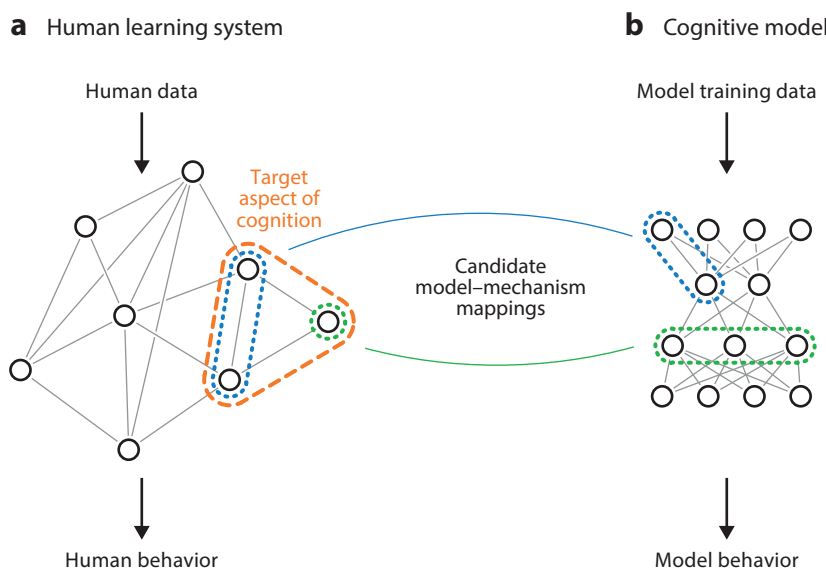


Figure 1

A schematic depiction of a model–mechanism mapping between (a) a human learning system and (b) a cognitive model. Candidate model–mechanism mappings are pictured as mapping between representations but can also be in terms of input data, architecture, or learning objective.

²In practice, it is not just one system (e.g., one person's perceptual system) but the class of such systems that humans—or humans of a particular age or culture, or all primates, or some other group—have. We will ignore this complexity for now; readers are referred to Cao & Yamins (2024) for discussion.

computational models are often complex artifacts whose relationship to the target human system of interest must be explicitly defined by the modeler. This correspondence between target system and model is sometimes referred to as a model–mechanism mapping (3M) (Cao & Yamins 2024, Kaplan & Craver 2011).

What counts precisely as a mechanism can be very controversial, and it often differs between cognitive science and neuroscience (Brun et al. 2025). For our purposes here, we simply mean that there is some mapping between the structure of the model and the structure of the target system, beyond an expectation that the behaviors of both should correspond. This mapping could be a correspondence between aspects of the model’s internal states and posited mental representations. It could also be a correspondence between specific aspects of the model’s training data and the human training data (abstracting over the model internals), or between model objectives and human motivational signals. We expect that many of these mappings can be productive; the key is that they are stated explicitly.

To make this discussion concrete, we consider the development of computational models of naive psychology.³ Naive psychology is, roughly speaking, the idea that people represent the beliefs and desires of other people and that they use these representations in reasoning about the actions they observe people taking (Premack & Woodruff 1978). These representations together constitute an intuitive theory in which inferences can be made about observed patterns of data from other agents (Chuey & Gweon 2025, Wellman 2018). Together, this reasoning process is dubbed a theory of mind, as it allows people to reason about the causes of others’ actions. Although initial theorizing about naive psychology was verbal, the complexity of this theory suggested that it would be important to formalize it in order to make more precise predictions.

Goodman et al. (2006) gave an early probabilistic cognitive model of naive psychology. In their model, probabilistic predictions could be made about observed actions on the basis of unseen entities such as beliefs and desires. This model made an explicit mechanism mapping: There were specific variables in the model that mapped directly to posited representations in the target system such as beliefs and goals. However, the model took in only the most schematic representations of the world: binary indicator variables representing the location of an object (0 = “in the box” versus 1 = “in the basket”) or the agent’s observed actions (0 = “searching in the box” versus “searching in the basket”).

Addressing this issue, more sophisticated probabilistic models of agent-based reasoning were developed that could make inferences about simple 2D blocks world scenarios in which schematic agents’ behaviors were observed (Baker et al. 2009, 2017). These models allowed quantitative predictions to be made about a wide range of scenarios and showed good correspondence with human judgments. They again contained explicit representations that were intended to map to the internal representations posited to be part of naive psychology, instantiating the theory that humans reason recursively about others as being rational, planning agents.

Despite the success of these models, there is no obvious route to generalize them to a broader range of stimuli. The format of the scenarios that they operate over is constrained by the specification of the model: The specific computations specified in the model only operate over 2D blocks worlds. The planning engine that the model uses for making judgments could in principle be applied to other discrete game-style worlds, but the effort involved in this adaptation would be considerable.

³Here we primarily rely on the term “naive psychology” to describe the broad set of ideas proposing that people represent an internal model of the minds of others. The term “theory of mind” is more commonly used, but it is overloaded because it can be used to refer to the capacity (i.e., making inferences based on belief) as well as a particular account of that capacity.

Stimulus Computable Models

None of this is new. Computational models of the type described above have been constructed to describe many aspects of human cognition, from perception to learning and decision-making, using symbolic representations, neural networks, and probabilistic models among other formalisms (Elman et al. 1996, Griffiths et al. 2024, Ma et al. 2023, Rumelhart et al. 1986). And there have been decades of debates about the inferences licensed by particular types of models (Jones & Love 2011, Pinker & Prince 1988). However, many of these questions take on new urgency because, unlike the previous generations of models, current models can be applied to a much wider variety of stimuli and situations.

We used the term “stimulus computable” above to describe this property of models. The term is not very precise, but it is meant to denote that something like the stimuli that humans experience in an experiment can be provided to a model in unmodified form. In some cases, this means providing precisely the same texts or images to the model as were given to human participants. In other cases, some compromise must be made—for example, in transcribing as text a spoken language stimulus, in providing a model with still frames as opposed to a video, or even in describing a visual stimulus in words. We discuss these compromises below.

In addition to being stimulus computable, there are a number of other features of modern AI models that together create a qualitative change in the cognitive modeling landscape:

- Modern AI models can succeed in a wider variety of tasks in perception and cognition, opening up new areas for study;
- Modern AI models can be trained on large amounts of realistic data—language, images, videos, or data in other modalities—making it feasible to simulate human learning and development at a scale unseen in prior work; and
- The accessibility of modern AI models—especially due to the ability to query AI models through chat interfaces and application programming interfaces (APIs)—means that they are being used by a wider range of researchers than ever before.

Yet, much is also still the same. When researchers use AI models as cognitive models, they must still make decisions about the scope and limits of the comparison between the human system and the computational system (Cichy & Kaiser 2019). For example, is the specific architecture type (e.g., a transformer) or learning rule (e.g., backpropagation) that a model uses important to the mapping, or are these critical details deemed irrelevant to the particular goals of a cognitive model? Is the runtime of the model intended to relate to the reaction time of humans? There is no easy answer to these questions: They are part of the messy work of debating the scientific use of specific models.

Modeling Naive Psychology Using Large Language Models

In contrast to the probabilistic inverse planning models described above, LLMs offer a flexible and powerful method of evaluating arbitrary stimuli, easily allowing them to be applied to a wide range of vignettes about naive psychology. An early paper took advantage of this opportunity to show that several commercial LLMs succeeded on standard probes of false belief understanding, which are often taken as a diagnostic for naive psychology more broadly (Kosinski 2024). These successes proved fragile and vignette specific (Ullman 2023), for some models at least.

In part inspired by this debate, a number of teams created large sets of vignettes probing different aspects of naive psychology that could be evaluated with both human participants and LLMs (Chen et al. 2024, Gandhi et al. 2023, Kim et al. 2023, Nickel et al. 2024, Shapira et al. 2024, Strachan et al. 2024). For example, Gandhi et al. (2023) used a highly capable LLM to facilitate

the generation of vignettes probing belief- and action-based inferences across a wide variety of contexts and then testing other LLMs using these vignettes.⁴

Yet, despite the inherent interest in evaluating these models, LLMs are not yet being used as cognitive models in the sense of the prior probabilistic models. In particular, there is no clear model–mechanism mapping that is established such that the correspondence between model and target system can be established. No states are identified as corresponding to beliefs or desires; no components of the model architecture are investigated as to their importance in producing successes. In short, no explanations of the behavior are posited. Instead, in this line of work, models are largely treated as black-box agents whose behavior can be evaluated only via the same techniques we use to evaluate human participants.

Creating a model–mechanism mapping for naive psychology is an important next step for this literature (Hu et al. 2025). For example, it might be productive to consider the input data required to produce psychological abstractions within a model, or to probe whether there are parallels between the abstractions that emerge in model states and those that are posited in humans. Given that interesting psychological reasoning currently emerges only in the largest models, forward progress will likely require solving a variety of technical challenges related to efficient model training and probing. Such steps might lead us closer to a true cognitive model of naive psychology.

MAPPING BETWEEN LEARNING MODELS AND HUMAN SYSTEMS

We argued above that computational models in cognitive science are limited mappings between some target part of human cognition and a formal artifact representing that computation. By virtue of comparisons between models, they can also permit inferences about what kinds of theoretical assumptions best match the human system. Yet the strength of these inferences is only as strong as the mapping between the model and the target system.

In modern, stimulus computable AI models, the mapping between model and human system usually comes down to two key issues: model training and model evaluation. We should not be absolutists about requiring a match on either of these dimensions between models and human systems. In some cases, a model can be informative even though its data or evaluation mismatches humans dramatically. To take our running example, Goodman et al.’s (2006) naive psychology model operated over highly schematic inputs. Precisely because of this simplicity, it showed that developmental changes in mentalistic reasoning could arise in highly simplified situations.

Model Training

As learning systems, LLMs and other recent AI models critically rely on their training data. To the extent that the modality, the distribution, or the quantity of the training data does not match the training data received by the target system (human learners), then this mismatch has the potential to undermine inferences from the model.

Data modality. If an LLM is being used as a cognitive model, this mapping first and foremost creates the assumption that language—and text in particular—is the appropriate learning input for the particular capacity being modeled. In some cases, this assumption might well be warranted (e.g., in modeling adults’ reading times; Wilcox et al. 2020). In other cases, training data that only

⁴Based on these experiments, our view is that the evidence is growing stronger for the presence of naive psychological reasoning within the most capable recent LLMs (cf. Hu et al. 2025). These models succeed on a wide variety of evaluations that span a range of social tasks and go beyond what might be present in their training data.

consist of text are likely less appropriate. For example, translating multimodal paradigms from child development research into prose could introduce substantial differences in the nature of the paradigm (Kosoy et al. 2023).

For models to provide a good description of other aspects of cognition, it is likely that text alone will not be sufficient. Increasingly, large commercial LLMs provide multimodal capabilities, though there is reason to suspect that these capabilities do not yet reflect the kind of visual cognition we see in humans (Schulze Buschoff et al. 2025). Some cognitive modeling work takes advantage of similar multimodal architectures. For example, an image-text model trained on data from human children was able to learn some links between words and images, providing a cognitive model for early associative word learning (Vong et al. 2024). Although this is an image-text model rather than a speech-video model, it still represents an important step forward in using multimodal learning for cognitive modeling.

Data distribution. Ideally, a cognitive model would be trained on the same kind of data that humans receive. Human children largely learn from grounded conversations that are appropriate to their developmental level (Huttenlocher et al. 2007), while adults' language diet is a mix of spoken and read text. In contrast, large pretrained LLMs are trained mostly on Internet text, which has no intrinsic curriculum or ordering. This mismatch of data distribution could be problematic for modeling child language learning or adult language processing (Feng et al. 2024, Hosseini et al. 2024).

A cognitive model could also be useful by mapping between features of the training data for humans and features of those given to models. In this case, the causal explanations enabled by the model would be about the properties of the data themselves—for example, that a particular data property would be required for a particular capacity to emerge. Several examples of this strategy come from investigations of reasoning behavior (Chan et al. 2022, Prystawski et al. 2023a). In one, researchers modified the higher-order statistics in the data distribution to show that only some distributions gave rise to the reasoning strategy called chain of thought reasoning (Prystawski et al. 2023a), in which LLMs can reason through complex problems by generating additional intermediate steps.

Examples from computer vision also show strong data distribution effects. Many of these models are trained on ImageNet, a large, curated database of images sampled uniformly across many different categories (Deng et al. 2009). This data distribution is radically different from that experienced by children (Huber et al. 2023). When computer vision models are trained on a more developmentally realistic distribution of visual data, they show substantially poorer performance, indicating a mismatch between their architectures and human learning architectures (Long et al. 2024b, Orhan & Lake 2024, Zhuang et al. 2021).

Data quantity. The remarkable successes of LLMs and other modern AI models have only been possible through the use of vastly more training data than humans receive (Frank 2023b). For example, while GPT-3—arguably the first model to show human-like in-context reasoning (Brown et al. 2020)—was trained on roughly 400 billion words of training data, human children likely receive fewer than 10 million words per year. Since in-context reasoning and other abilities emerge relatively early in children, there is thus a likely data gap of many orders of magnitude between the training data necessary for training AI models and the input necessary for children (**Figure 2a**). Thus, data quantity appears to be a key disanalogy in the use of AI models as cognitive models.

Yet to make this data gap more precise, we need to measure model and human performance in comparable ways so that human development can be compared with model scaling (**Figure 2b**; see discussion of evaluation in the next section). The apparent pattern for some aspects of cognition

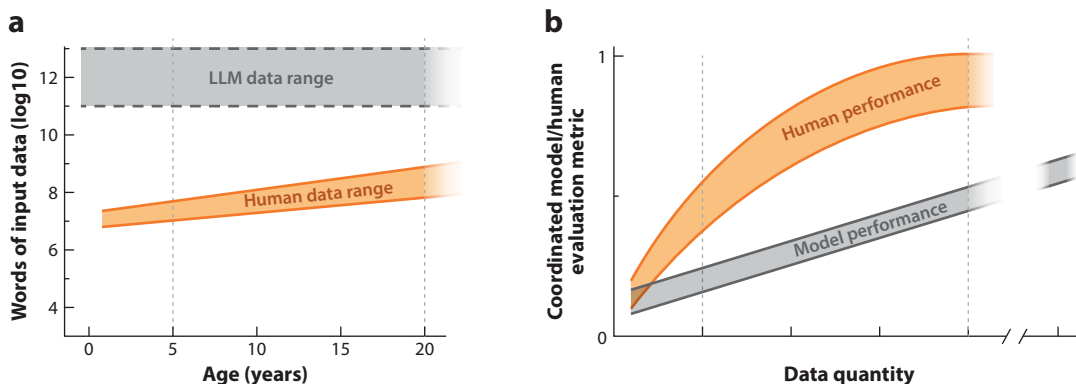


Figure 2

Data efficiency in human learning. (a) Order of magnitude of large language model (LLM) versus human training data, plotted by human age. Ranges are approximated from Frank (2023b). (b) A schematic depiction of evaluation scaling curves for human learners versus models, plotted by training data quantity.

is that model performance scales with data much more slowly than human performance, but this hypothesis must be tested quantitatively across domains.

The BabyLM challenge is a notable case in which natural language processing researchers have pursued this type of alignment. BabyLM is a machine learning competition in which groups compete to train AI models on a small, fixed data set (canonically, 100 million words) in order to explore what can emerge from human-scale training corpora. The systems that are submitted as entries to the BabyLM challenge are typically not posed as cognitive models, but they nevertheless lay the groundwork for systematic comparisons of cognitive models trained on the same scale of data as human agents (Hu et al. 2024b, Warstadt et al. 2023).

Model Evaluation

Cognitive models are evaluated based on their relationship to target phenomena, but the nature of this evaluation can vary widely. The simplest such method is to identify qualitative patterns in human behavior and ask if they emerge in models. One simple and elegant probabilistic model produced an exponential increase in vocabulary size across word learning based on the absolute simplest set of assumptions possible; this demonstration suggested that more complex assumptions were unnecessary (McMurray 2007). Stimulus computable models from modern AI models offer far richer quantitative options for evaluation, while also creating many more opportunities for mismatch between human and model evaluation.

For example, it is possible to ask LLMs to produce explicit grammaticality judgments on sentences (e.g., “Is the following sentence grammatically correct in English?”) and to compare these judgments to the judgments of human participants. In one such comparison, model and human judgements did not match, potentially suggesting that LLMs should not be used as theoretical tools for investigating human language processing (Dentella et al. 2023). The trouble with this inference appears to be connected to the method used for model evaluation, however. When grammaticality was assessed using comparisons between minimal pairs of sentences (one grammatical and one ungrammatical), this metric yielded a strong fit to human acceptability judgments (Hu et al. 2024a). This example illustrates one of the wide variety of challenges that arise in cognitive model evaluation.

Below we describe a set of principles for model evaluation, drawing on prior collections (Frank 2023a, Ivanova 2024) as well as on general principles of experimental design for human experiments (Frank et al. 2025). The general approach we take here is that the methods of “machine psychology” should often follow best practices for research with humans in the areas of measurement, experimental design, and stimulus creation.

Consider how measures are derived from models. The way we interrogate human behaviors matters to the answers we receive. Thus, the development of appropriate measurement techniques for human participants is an important focus of research methods development in psychology (Frank et al. 2025). In the same way, evaluations of cognitive models require researchers to think about measurement. As we have discussed, in many experiments LLMs can be made to respond in the same format as humans.

Although using prompts to ask models to generate an explicit judgment can be a powerful method for aligning human and model tasks, this method can also have substantial weaknesses. Just as human explicit and implicit judgments are not always aligned (Nosek 2007), prompt-based judgments from models can sometimes underestimate the models’ latent knowledge. Indeed, examination of raw next-word probabilities often provides a more accurate measure of what LLMs “know” (Hu & Levy 2023). This issue is especially salient in cognitive models, which are more likely to be trained on smaller, more realistic data sets: The mismatch between prompt-based versus probability-based evaluations tends to be larger in less capable models (Hu & Frank 2024).

Researchers must not only derive a particular measure from models but also compare it to other measures. This comparison requires a linking hypothesis, that is, a hypothesis about how model-derived metrics correspond to human decision (Shah & Varma 2025). Common linking hypotheses in cognitive modeling include the use of softmax choice rules to simulate a forced choice between alternatives (Luce 1977) and the use of surprisal to simulate reaction times (Levy 2008). In contrast, the presence of prompt-based evaluations allows for a seemingly transparent linking hypothesis: The models’ responses are simply correlated with human judgments. This strategy may not be optimal, however. Experiments often measure human participants based on forced choice judgments or implicit measures such as reaction time rather than by asking them to perform open-ended language generation tasks, and our models should predict these dependent measures as well (Wang et al. 2024).

Use experimental designs to make causal inferences. Machine learning evaluations typically focus on models’ performance on a particular task, but standard benchmarking practices do not allow for scientific inferences. The general logic is that benchmarks represent a difficult task, and performance on them is indicative of a model’s level of capability. Often multiple models are evaluated on the task, but it is not possible to make causal inferences about what makes one model perform differently than another, given that major model families typically differ in their architecture, training data, and size.

Effective cognitive modeling requires experimental designs that leverage specific manipulations to make causal inferences. Controlled contrasts between variants of the same model (sometimes called ablation experiments) are quite common in machine learning, where the focus is often on model training and comparison. In contrast, in the rush to adopt LLMs for exploring cognitive modeling, technical obstacles around retraining models have sometimes prevented researchers from running appropriate experiments.

A second kind of controlled experiment is a contrast between different stimulus conditions. In standard benchmarks there are often different classes of examples; for example, many of the intuitive psychology benchmarks discussed above include examples from different domains of social and pragmatic reasoning. Comparison between these conditions can be interesting from a

descriptive perspective, but the examples often differ on many dimensions. For example, Strachan et al. (2024) used different vignettes with different lengths and characteristics to evaluate irony compared with false belief comprehension. These differences confound the comparison between conditions, so it cannot be concluded that models necessarily differ on these capacities just because their scores differ. In contrast, benchmarks afford stronger inferences if they make minimal changes between pairs of conditions (Gandhi et al. 2023).

Assess the validity of measures. In psychology, the validity of a measure is the degree to which it measures the construct it is supposed to measure (Borsboom et al. 2004, Frank et al. 2025). For example, a naive psychology benchmark that also involved solving mathematical equations (“What does Sally believe is the solution to the equation in the basket?”) would be measuring not only the model’s naive psychology abilities. Unfortunately, threats to validity are not always this straightforward, as the structure of the constructs being measured is often unknown. The discussion of naive psychology above is a case in point: If two assessments that ostensibly measure social reasoning do not group together across different models, this result could indicate issues in one or both of the assessments or differences in how the construct itself is structured in the models. In light of this ambiguity (which also exists in studying humans!), the only recourse of the careful investigator is to attempt to find converging methods to study the robustness of a theoretical claim.

Threats to validity are not the same for all models. Smaller and less capable models—precisely those that can be useful for cognitive modeling—are more likely to be affected by other aspects of a benchmark’s structure. For example, smaller models may perform worse on tests of reasoning or language comprehension simply because of the complexity of a task prompt (Hu & Frank 2024).

One particular validity concern deserves special mention: data contamination. Any experimental evaluation must avoid reusing materials that are likely to be in the training data, since any description of these materials (for example, in a scientific paper) will likely be accompanied by normative or descriptive answers that would bias the evaluation of the model. Given the vast amount of data used by pretrained language models, this risk is very real: Almost any stimulus found in the published literature is likely to be present in training corpora (Deng et al. 2024). The key question is how much the available materials must be modified to provide a true test of generalization. The literature on naive psychology discussed above is informative here: Relatively closer generalizations of vignettes found in the existing literature tend to produce successes, while relatively more creative or novel problems tend to produce more mixed results (Hu et al. 2025). To the extent that a model’s responses depend on the specific questions (or very similar ones) being present in the training data, the validity of that assessment is compromised.

OTHER THEORETICAL CONTRIBUTIONS OF AI MODELS

Not every AI model is a cognitive model; most are not. While we have argued that cognitive models play a privileged role in advancing our theoretical understanding, AI can be helpful in many other ways, some of which may serve as stepping stones to cognitive models. In this section we describe three of these contributions (cf. Ong 2024 for a related taxonomy), considering how models can be used as machine research assistants, as human simulators, and as examples of other minds.

Machine Research Assistants

Given their capacity for image and text processing as well as other interactive capacities, AI systems hold immense promise for enhancing research into human cognition by acting as machine research assistants, that is, by replacing the use of human effort to categorize and annotate images, drawings,

or language productions. Supplanting another contribution of human effort, they can also be used as machine confederates in experimental studies, allowing controlled interactions as part of an experimental protocol.

Demszky et al. (2023) provide a review of some of the possibilities afforded by such models (see also Ke et al. 2025). On the side of stimulus generation, generative AI models can be used to create both verbal and visual stimuli at scales that would otherwise be unwieldy or cost prohibitive, such as, for example, creating illustrations or vignettes. AI models can also be used in analyzing text, image, or video data, for example, by extracting the features of drawings (Long et al. 2024a), computing the similarity of conversations (Boyce et al. 2024), or extracting the locations of faces (Long et al. 2022). These tasks would often require hundreds or thousands of hours of human effort and thus would be practically impossible without model assistance. Yet models—just like human annotators—are imperfect instruments and thus care must be taken to assess and communicate their accuracy on annotation tasks (Ziems et al. 2024). Interrater reliability (between humans and also between models) and manipulation checks within generated stimuli are two methods that relate to the construct validity efforts described above.

The use of models as confederates can also create opportunities for new studies of human cognition, though just as in conventional confederate research, researchers must consider the risks associated with deceiving participants about the nature of their partners (Leavitt et al. 2021). Often with AI confederates, intervention is explicitly marked as coming from an AI model. For example, language models can generate consensus statements summarizing group discussions of controversial political issues (Tessler et al. 2024) or even reduce beliefs in conspiracies (Costello et al. 2024).

Human Simulators

Because AI models are stimulus computable, they can be used to predict human responses to experimental items, even if there is no specific theory to explain why they do so. Initial experiments of this type showed that an LLM (GPT-3.5) reproduced human moral judgments with surprising accuracy (Dillion et al. 2023), raising the possibility that models could be substituted for human participants in some applications (for example, for pilot testing new experimental materials). Since then, this technique has proven successful across a range of domains from social science to neuroscience (Aher et al. 2023, Argyle et al. 2023, Lippert et al. 2024, Luo et al. 2025). We argued above that models are important for both prediction and explanation, whereas human simulators are primarily focused on prediction alone.

Pure prediction of human behavior can nevertheless have important implications. For example, experimental effect sizes can be predicted from nationally representative survey experiments based on evaluations of their stimuli (Hewitt et al. 2024). This result opens up many new avenues for identifying successful experimental interventions more efficiently—for example, by conducting *in silico* pilot evaluations and then testing only those interventions that are predicted to yield large effects.

Prediction of human responses can also be a stepping stone toward cognitive modeling. For example, Centaur is a pretrained LLM that is fine-tuned on trial-by-trial data from a large corpus of cognitive psychology studies (Binz et al. 2025). Although the base model is already impressive in predicting participant choices, the fine-tuned model achieves even better accuracy on held-out trials, handily outperforming domain-specific cognitive models from the literature. The authors argue that accurate prediction of this type is the first part of creating a unifying model of human cognition. Toward this claim, they show that model representations become more aligned with human neural representations after fine-tuning on the behavioral experiment data. These results

are exciting; yet it also seems intuitively surprising that a model of human cognition might begin with an artifact that is pretrained on 15 trillion tokens of data (Meta AI Research 2024), essentially incorporating an approximation of the full cultural production of billions of individuals.

Human simulators are both pragmatically useful and theoretically interesting, but despite their impressive predictive successes, they provide limited explanatory traction. Deepening these systems will require layering on further mappings to the human system—to representations, architectures, or even data—beyond a simple match in behavior.

Other Minds

Comparative studies of animal intelligence have long provoked reflection on what it means to be intelligent. For example, octopuses show remarkable adaptive behaviors emerging from a distributed system within an individual organism (Godfrey-Smith 2016), while ants and other social insects show intelligent behavior at the level of a network of organisms (Gordon 2010). These examples suggest that centralized processes of reasoning and cognition need not be the only way that adaptive problem-solving can emerge, broadening our appreciation of the diversity of routes to intelligence.

AI can serve the same purpose. Long prior to the most recent innovations, cognitive scientists have been inspired to think about human intelligence in terms of, and in comparison with, technical artifacts from telegraphs to computers (Kirkland 2002). The success of engines for playing games such as chess (Campbell et al. 2002) or checkers (Schaeffer et al. 2007) often prompts reflection on how these systems, as impressive as they are, show limited resemblance in their mechanisms to how humans are thought to play the games. Like the comparative biology examples above, they again suggest that seemingly intelligent behaviors can be produced by very different kinds of algorithms. The strongest possible outcome of comparisons across the space of intelligences would be universal hypotheses about intelligent systems.

Here we discuss two lines of this sort of research. The first consists of broad case studies of convergence in intelligent behavior. The second is the special case in which models provide a proof of concept that some behavior is in principle learnable for some agent, regardless of its internals.

Comparative case studies. The idea of using general solutions to difficult problems as a constraint on our hypotheses about cognition has a long history in cognitive science: Marr (1982) described analysis of the problems facing human minds and brains at a computational level, in which the analyst considers only how a problem can be solved given the information at hand. Similarly, Simon suggested that the primary route to a science of AI would be to characterize the workings of an agent by the interfaces between its environment and its goals, irrespective of the precise mechanisms that carry out a computation (Simon 2019). The idea in each of these cases is that, if an organism solves a particular hard problem, you can learn about how this happens by examining any system smart enough to solve that problem. Indeed, perhaps as organisms or systems become more intelligent, they converge on similar representations (Huh et al. 2024).

One area in which this parallel is being advanced is the study of formal and mathematical reasoning in LLMs. In the absence of specialized training, large pretrained language models generally tend to struggle on tests of mathematical reasoning (Imani et al. 2023, Lewkowycz et al. 2022). Yet a variety of recent innovations led to improved performance on math tasks as well as other reasoning assessments (DeepSeek-AI et al. 2025, Wei et al. 2022, Zelikman et al. 2022). These reasoning models are already being compared to humans (Liu et al. 2025, Prystawski et al. 2023b). For example, strategies that are helpful in facilitating reasoning for both language models and humans are more likely to be general than those found only in humans (Gandhi et al. 2025, Prystawski et al. 2023a). It is likely too soon to tell whether reasoning models make good cognitive models;

they are *prima facie* quite dissimilar in the type and amount of training data they receive. Yet, as our most capable approaches to modeling reasoning, they are—like octopuses—very interesting prompts for speculation about the nature of intelligence.

Proof-of-concept learners. One special case of using AI models as other minds comes from their application to learnability debates. In linguistics especially, there has been immense historical interest in questions about what representations can be learned from human linguistic input. The “argument from the poverty of the stimulus” is the argument that certain aspects of natural language structure—grammatical structures especially—are not learnable from the language without some kind of innate assumptions or biases (Laurence & Margolis 2001). This argument was supported by early formal results showing that natural languages were formally unlearnable under restrictive, adversarial assumptions about the nature of the learning context (see Clark & Lappin 2010 for review).

Since modern LLMs have only relatively flexible and general assumptions about the nature of the data they observe, their success in producing grammatical language is a *prima facie* challenge to stimulus poverty arguments (Piantadosi 2024). This challenge does not rest on any particular mapping of LLMs to human cognition, though: If language can be learned by any system, then it is in principle learnable!

Beyond the broad success of models in generating grammatical language, LLMs can also be used for more specific proof-of-concept learning demonstrations. For example, special-purpose LLMs can be trained on corpora that lack specific constructions so that the model’s judgments about these constructions can later be probed (Misra & Mahowald 2024, Warstadt & Bowman 2020). Alternatively, models can be compared on their learning of human languages versus impossible variants of these languages, testing whether possible human languages have structures that make them generally easier to learn (Kallini et al. 2024, Xu et al. 2025, Yang et al. 2025). These demonstrations creatively exploit the general learning prowess of models to ask questions about learnability in principle, rather than in a specific human system.

Some responses to this general set of challenges suggest that these in-principle arguments are not compelling, because LLMs are poor cognitive models of language learning (Katzir 2023, Kodner et al. 2023). Regardless, they defeat prior in-principle learnability arguments. Moreover, others (including us) are more positive as to the potential for cognitive modeling of language using LLMs (Futrell & Mahowald 2025). In-principle demonstrations can often be a first step toward tests of more specific hypotheses about how humans solve difficult problems.

CHALLENGES AND OPPORTUNITIES

We end by considering some of the challenges and opportunities facing the field of cognitive modeling in the age of new stimulus computable models, including reproducibility, generalizability, and interpretability.

Transparency and Reproducibility

Scientific theories cannot be secrets; for a theory to be accepted by the scientific community, it must be shared across that community. When a theory consists of a small number of equations or a verbal description, this statement is tautological, but when a theoretical artifact consists of an architecture that is specified in code and then trained on a data set of many millions or billions of tokens, transparency and accessibility are far from assured.

The key property for theoretical artifacts is reproducibility. Reproducibility means that, for a given result, independent investigators must be able to carry out the same computations as the original investigators and observe the same result; this property is sometimes referred

to as computational reproducibility. Despite the foundational nature of reproducibility, it is by no means assured. Even for simple experimental data analysis, many numerical results are often irreproducible (Hardwicke et al. 2018, Nuijten et al. 2016); in more complex pipelines, reproducibility is even more limited (Stodden et al. 2018). Reproducibility problems for the complex architectures used in AI and machine learning are severe (Haibe-Kains et al. 2020). For such work to be reproducible in principle—even if true reproducibility may be out of reach due to constraints on computation resources—researchers must share code, data, and details of the computational environment in which training was completed (Stodden et al. 2016).

A profusion of experimental studies, including many reviewed above, have compared human behavior to models in the GPT family, which have included many of the most capable available models. While many of these experiments are interesting, models like GPT-3, GPT-4, and their successors are not suitable for being used to build theories (Frank 2023c). Due to their development as industrial products, key features of their architecture are unknown; their training data and the details of their training are private; and they are only accessible through Application Programming Interface (API) endpoints. Thus, there is no guarantee that any particular result using these models can be reproduced in the future.

The rise of open access LLMs means that far more models are available with open code, open neural network weights, and open training data (Kukreja et al. 2024, Liesenfeld et al. 2023). Despite the convenience of commercial APIs, it is thus critical that researchers interested in the scientific use of AI models choose these open models.

Population Generalizability

Theories in psychology and cognitive science must explain not only consistency between individuals but also variability. Indeed, the scope and limits of this variability are often thought to be key features for theories to predict (Frank et al. 2021, Piantadosi & Gibson 2014, Underwood 1975). Yet the vast majority of cognitive modeling—both in prior traditions and using current AI models—has focused on predicting average patterns of human performance. These averages can be especially problematic as a focus because they are typically taken from (online) convenience samples, meaning they are not in fact estimates for any particular population (Doebel & Frank 2024, Landers & Behrend 2015, Sherman 2024).

A match between AI models and human data must be interpreted in light of the source of both the human data and the model's training data. Most language data for training LLMs are sourced from the Internet, which vastly overrepresents particular languages, cultures, topics, and modes of writing. These sampling biases in training data lead to models that represent the knowledge and attitudes of a subset of human experiences: in particular, those of English-speakers with Internet access (Atari et al. 2023).

Much more work is necessary to understand how well models can reflect variability across individuals and across cultures (Abdurahman et al. 2024, Agnew et al. 2024). One promising line of research attempts to fine-tune models based on in-depth language data produced by individuals from a diverse range of backgrounds (Park et al. 2024). This approach yields models that act as better human simulators; however, each fine-tuned model artifact still contains an entire Internet's worth of training data. It remains to be seen whether similar approaches can be taken for cognitive models trained on human-scale data from variable sources (Murthy et al. 2024).

Interpretability

A classic challenge to the use of neural networks as cognitive models is the assertion that they are black boxes, whose utility is limited by the inability to interpret their internal states

(Bowers et al. 2022, Kay 2018, Marcus 2018). Since most modern models have at least millions (and often billions) of parameters, exploratory data analysis for these representations is much more challenging than it was in the first wave of connectionism (e.g., Hanson & Burr 1990). Yet a profusion of new tools allows cognitive modelers to make connections between internal representations and posited mechanisms (McGrath et al. 2024). These tools allow for mapping of models to neural representations and, increasingly, for the creation of mappings to cognitive representations as well.

Internal model representations can be used to predict neural representations. Early work using convolutional neural networks for vision showed that models that performed better on image categorization also had representations that matched human functional activation (Yamins et al. 2014). This general result has now been replicated across a variety of different neural modalities and domains—including vision, audition, and language (Hosseini et al. 2024, Kell et al. 2018, Schrimpf et al. 2021, Zhuang et al. 2021). Indeed, the mapping between network states and neural data is so well-established that there is a leaderboard for neural prediction and models can be entered in domain-specific competitions for the best prediction of new data (Schrimpf et al. 2020).

Increasingly, new mechanistic interpretability tools can be used for creating cognitive mappings as well, making it easier to make useful model–mechanism mappings even in complex models. For example, research into linguistic syntax has used such tools to probe for the presence of syntactic categories and structures (Manning et al. 2020, Tenney et al. 2019). These techniques show promise in helping to understand the mapping between traditional symbolic abstractions and the representations learned by models (Linzen & Baroni 2021, Pavlick 2022). Newer techniques such as sparse autoencoders can even be used to discover cognitively interpretable structures in very large language models (e.g., Huben et al. 2024).

Interpretability techniques typically work by identifying the presence of some structure in model representations. Given the vast number of parameters for most modern networks, perhaps it is not surprising that information relating to certain representations is present in some form, but it could very easily be present and not play any causal role in the network’s computation. To address this issue, causal probing techniques allow for the computational role of specific representations to be interrogated (Geiger et al. 2021, 2024). For example, these techniques can be used to ask which of several hypothetical symbolic computations a very large language model approximates in its computations for a particular task (Wu et al. 2023). The application of these techniques to cognitive models provides an exciting potential tool for understanding the nature of the computations that networks carry out (Geiger et al. 2023, 2024).

CONCLUSION

Progress in cognitive science requires theories that can explain and predict human behavior. Computational models are our best tools to instantiate these theories explicitly and connect them to data. Because our behavior is diverse, variable, and environmentally conditioned, such models must operate over a wide range of stimuli and training data. Because of the flexibility and power of modern AI models, they open up many exciting opportunities to create stimulus computable cognitive models—models that can make predictions about human behavior across an arbitrarily large and realistic set of stimuli.

Taking advantage of this opportunity requires careful thinking about what makes an AI model a scientific model. We have argued here that a thoughtful model–mechanism mapping is the first step, in which the training and evaluation of a model are aligned with a particular aspect of human cognition. As we stressed above, the intent of this alignment is not to provide an exact match; some of the best models are substantial simplifications of their target system. However, the validity of a model rests on an evaluation of this alignment: If the assumptions of the mapping

are incorrect, then predictions from the model will be incorrect as well, and theoretical inferences will be undermined.

This is an exciting time. Many of the dreams envisioned by the founders of cognitive science have become reality. As those founders envisioned, the fusion of AI with psychology and linguistics yields the potential for new insights into the nature of the human mind.

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